Software Test Plan

Traffic Management System (Simulation)

Document control	Value
Course	Software Engineering UE24CS341A
Institution	PESU
Document edition	1.1 - consolidated to Baseline Amendment v1.1
Normative baseline	Plan edition 1.1; tests trace to SRS baseline 1.0
Revision date	6 November 2026
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Status	Consolidated for submission; approval claims limited to recorded team decision

Revision history

Edition	Date	Change	Approval
0.2	2026-09-07	Simplified SRS finalized for implementation and review.	Pending in original record
0.3	2026-09-14	SAD created for TMS-Sim.	Pending in original record
1.0	2026-09-15	Software Test Plan created from the SRS and SAD.	Pending in original record
1.1	6 November 2026	Consolidated edition incorporating Baseline Amendment v1.1; does not claim new faculty approval.	No new approval claimed

Course Coordinator approval is not recorded in the source documents reviewed.

1 Introduction

1.1 Purpose and scope

This plan defines testing scope, strategy, environment, traceability, entry/exit criteria, schedule, responsibilities and reporting for TMS-Sim. Testing covers road-network modelling, fixed-time signals, demand, fixed-step simulation and controls, 2D visualization, metrics, persistence, comparison, CSV export and the local scenario library.

1.2 Exclusions and references

Out of scope: real signals/sensors/field hardware, 3D or distributed simulation, advanced lane changing, field calibration, public transport and pedestrian/cyclist micro-simulation.

- SRS_v1.1 and SAD_v1.1, consolidated to Baseline Amendment v1.1.
- docs/rtm_draft.csv is the current requirement-to-test/status source; docs/test_catalogue.md contains test definitions.
- docs/test_execution_record.md and artifacts/ record actual execution evidence, not plan criteria.
- DEVIATIONS.md and docs/issue10_acceptance.md record implementation differences and metric conventions.

2 Test items

- Network editor and validation
- Fixed-time signal configuration and validation
- Traffic demand generation and seed reproducibility
- Simulation engine, duration and controls
- 2D visualization
- Metrics, dashboard, run persistence, comparison and CSV export
- Local scenario library

3 Features to test and traceability

All retained test IDs below match docs/test_catalogue.md and docs/rtm_draft.csv. A mapping or aggregate suite result does not establish that an individual case passed.

Test ID	Requirement	Planned verification
TC-NET-01	TMS-F-010	Create/edit directed network; save/reload attributes.
TC-NET-02	TMS-F-011	Configure control types and permitted turning movements.
TC-NET-03	TMS-F-012	Test disconnected graph, zero length, zero lanes and missing receiving lane; block invalid run.
TC-NET-04	TMS-F-013	Import/export complete version-tagged JSON scenario in clean store.
TC-SIG-01	TMS-F-020	Define ordered fixed-time phases and green/yellow/all-red durations.
TC-SIG-02	TMS-F-022	Reject invalid timing and conflicting green movements.
TC-DEM-01	TMS-F-030	Configure entry rates and turning ratios; compare generated counts with a test-stated tolerance.
TC-DEM-02	TMS-F-032	Repeat scenario and seed; compare arrivals and metrics for determinism.
TC-SIM-01	TMS-F-040	Advance fixed steps and verify model and signal behavior.
TC-SIM-02	TMS-F-041	Start, pause, step, reset; verify 0.5x, 1x, 2x and 4x.
TC-SIM-03	TMS-F-042	Complete configured duration; verify final metrics persist once.

Test ID	Requirement	Planned verification
TC-SIM-04	TMS-F-043	Compare rendered state to engine; verify pan and zoom.
TC-MET-01	TMS-F-050	Independently recompute lifecycle/measurement metrics, including empty and incomplete runs.
TC-MET-02	TMS-F-051	Verify live updates and final dashboard values.
TC-MET-03	TMS-F-052	Round-trip scenario snapshot, seed, configuration, metadata and metrics.
TC-MET-04	TMS-F-053	Compare persisted runs including identity, units, unavailable values and compatibility.
TC-MET-05	TMS-F-054	Parse CSV; verify stable keys, labels, units, values and error handling.
TC-LIB-01	TMS-F-060	Save, list, load, rename and delete in isolated store.
TC-PERF-01	TMS-NF-001	Reference scenario, seed 42, 1x, one simulated hour; measure complete-run average FPS.
TC-REL-01	TMS-NF-002	Three clean one-hour runs; record configuration and result for each.
TC-UX-01	TMS-NF-003	Moderated new-user test; record participants, tasks, timing and assistance.
TC-PORT-01	TMS-NF-004	Clean Windows 10 and 11 runs; record OS/build and setup.

4 Test approach

Level/type	Application
Unit	Isolated simulation, network, demand, metrics, storage, reporting and UI modules.
Integration	Validation, demand, engine, metrics, storage and report interactions.
System	Analyst workflow from setup through simulation, results, persistence, comparison and CSV.
Acceptance	Assess retained SRS criteria using RTM and approved baseline.
Regression	Re-run affected cases after changes.
Performance	Full one-hour reference run using the defined FPS measurement.
Usability	Moderated new-user testing; automated workflow is not a human test.
Reproducibility	Same scenario/seed, compare arrivals and metrics.

4.1 Security validation

Validate safe local input/storage: oversized, malformed and wrong-version scenario files are rejected before parsing or storing scenario data. Verify normal operation does not require network communication. Login, role, administration and audit-log testing are out of scope.

5 Entry, exit and suspension

5.1 Entry criteria

- Stable build and required fixtures/data are available.
- Windows/runtime/dependencies required by the test are available.
- Tests use isolated temporary databases/stores; they do not write to normal user run-history data.

5.2 Exit criteria

- Planned cases are executed or explicitly marked blocked/not run.
- High-priority functional requirements pass mapped tests; all NFRs have evidence against approved criteria; TMS-F-032 passes.
- RTM links retained requirements to tests and evidence; no open Critical/High defect remains.
- Record pytest assertions and process exit separately; a nonzero exit is not a clean process pass even when assertions passed.
- Missing human/platform evidence and ambiguous criteria are blockers, not passes.

5.3 Suspension and resumption

Suspend if the application cannot start, the environment is unavailable, or a blocking defect prevents a significant portion of testing. Resume once the defect is fixed, a stable build is available and the environment/data are restored.

6 Test environment and data

Area	Baseline
Performance hardware	4-core Intel Core i5-11400H at 2.70 GHz, 16 GB RAM; standard display, keyboard and pointing device.
OS	Windows 10/11; record exact edition/build and clean-install status.
Runtime	Python 3.11 + PySide6; stdlib sqlite3; versioned JSON and CSV.
Tools	pytest; GitHub Issues and GitHub Actions on Windows/Python 3.11. Jira is replaced.
Test data	Valid/invalid network/signal scenarios, demand settings, seeded runs and performance_10_intersections.json.
Storage safety	pytest temporary paths and isolated history databases; no silent fallback to user data.

7 Schedule

Execution follows sprints: Sprint 1 ran 19-27 October 2026; Sprint 2 runs 27 October-6 November 2026. Acceptance evidence is in docs/test_execution_record.md and artifacts/. These dates are schedule context, not evidence that a case was run.

8 Deliverables

- Software Test Plan and test-case catalogue.
- Fixtures/sample scenarios; test execution record and summary.
- Defect log, RTM updates, performance/reliability evidence and usability/portability evidence when executed.

9 Roles and responsibilities

Member	Responsibility
Venkata Srivaibhav Nukaraju (PES2UG24CS579)	Team Lead; simulation engine and core model
V Vyas (PES2UG24CS569)	Network and signal-plan modelling
Vikas N Naik (PES2UG24CS583)	UI, 2D visualization and UX
Vikhyath Bharadwaj K S (PES2UG24CS586)	Metrics, reporting, QA and test

10 Risks and mitigation

Risk	Mitigation
Simulation defects affect system testing	Run unit/integration tests first; debug with small scenarios.
Large scenarios reduce rendering rate	Use reference scenario and 15 FPS target on reference hardware.

Risk	Mitigation
Invalid network/signal input	Maintain positive and negative tests from SRS validation rules.
Scenario data loss/corruption	Use controlled copies; verify supported storage/import/export paths.
Native shutdown/environment failure	Preserve diagnostics; distinguish assertion summary from process exit.

11 Assumptions and unresolved test detail

- Users may supply data or use bundled scenarios; correctness is against the documented model, not field measurements.
- TC-REL-01 follows approved wording of three clean one-hour runs. The amendment does not define 'clean', GUI/headless mode, seed allocation or interruption handling.
- TC-UX-01 is a moderated new-user test; no participant count, task threshold, time limit or assistance rule is specified.
- TC-PORT-01 requires clean Windows 10 and Windows 11 runs; exact builds and clean-install procedure are unspecified.
- Clarify these gaps in controlled baseline records before using them to judge pass/fail when the interpretation changes the result.

12 Test case management and traceability

The current RTM (docs/rtm_draft.csv) supersedes STP v1.0 §13. It includes all mappings below, including amendment additions TMS-F-011/TC-NET-02, F-013/TC-NET-04, F-042/TC-SIM-03, F-051/TC-MET-02 and F-052/TC-MET-03. TMS-F-021 and TMS-F-031 are retired and have no test mapping.

Requirement	Test
TMS-F-010	TC-NET-01
TMS-F-011	TC-NET-02
TMS-F-012	TC-NET-03
TMS-F-013	TC-NET-04
TMS-F-020	TC-SIG-01
TMS-F-022	TC-SIG-02
TMS-F-030	TC-DEM-01
TMS-F-032	TC-DEM-02
TMS-F-040	TC-SIM-01
TMS-F-041	TC-SIM-02
TMS-F-042	TC-SIM-03
TMS-F-043	TC-SIM-04
TMS-F-050	TC-MET-01
TMS-F-051	TC-MET-02
TMS-F-052	TC-MET-03
TMS-F-053	TC-MET-04
TMS-F-054	TC-MET-05
TMS-F-060	TC-LIB-01
TMS-NF-001	TC-PERF-01
TMS-NF-002	TC-REL-01
TMS-NF-003	TC-UX-01
TMS-NF-004	TC-PORT-01

13 Metrics and reporting

- For each case record status (pass/fail/blocked/not run), date, commit, environment, command, duration, evidence path and defect link. Associate each automated result row with its pytest process exit status from CI.

- For pytest, record assertion result and process exit status from CI separately, plus failed/skipped counts and available traceback/shutdown details.
- Track executed/pass/fail/blocked, requirement coverage, defect count/aging, performance FPS and reproducibility.
- Do not infer an individual catalogue test passed solely from an aggregate suite result.

14 Approval and control

The Issue #1 decision record states all four teammates approved baseline decisions on 27 October 2026. Original course-coordinator approval fields are blank; no faculty approval is claimed. Requirement, criterion or interface changes require controlled baseline review and RTM/STP updates. Original Part 1 PDFs remain unchanged.